GeoScope | Definition-aware atomistic evidence verification

One physical-proof criterion is tested under complementary source and evaluation conditions.

A UMA-relaxed OC20-Dense

55 qualified source groups | 1,760 claim variants

3 LLMs; arithmetic / comparison-ID / full checks

B OC20 stored DFT trajectories

48 qualified source groups | 1,536 claims per model

4 LLMs; strict / normalized / operation diagnostic

Shared physical evidence + licensed-proof rules

Atomic endpoints | atom subsets | frame / alignment | periodic treatment | weights

Measurements M, A, I | permitted numerical operations | supported / contradicted / insufficient

Proof-validity assessment

What weaker checks miss; verdicts and coverage

Definition-aware checks reject unsupported selections

Robustness and reliability diagnostics

Source variation; format; evidence; operation

Abstention and decision-token confidence

No claim of closed-loop catalyst discovery or a hybrid advantage over deterministic-only proof enumeration.